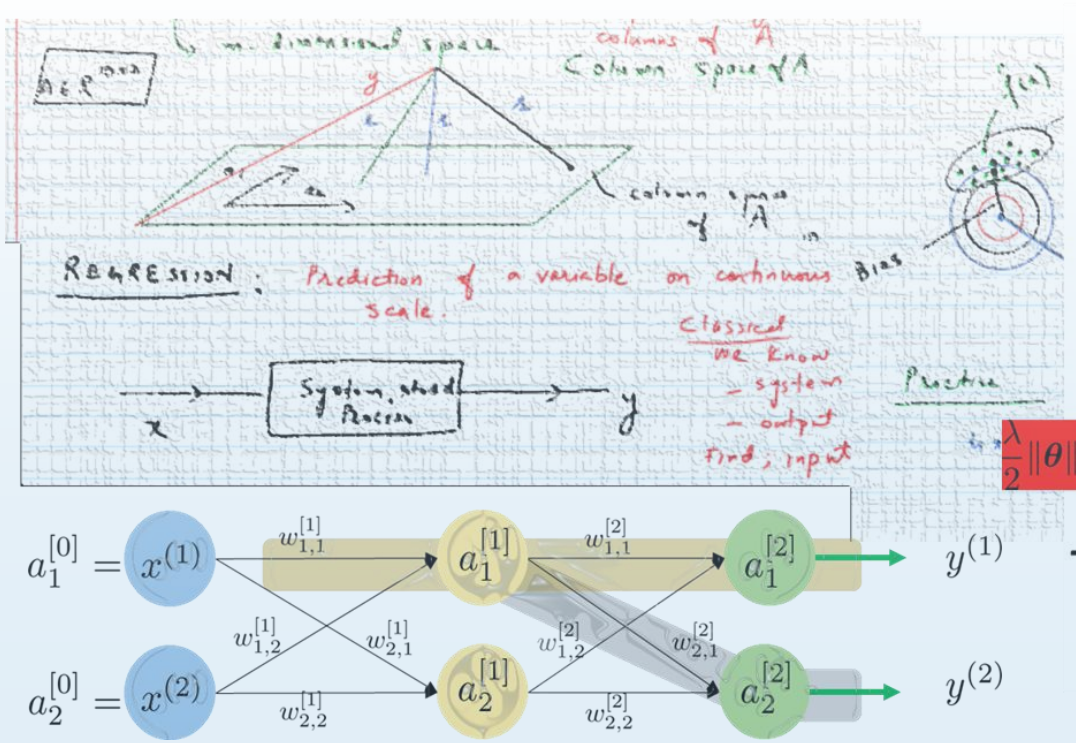
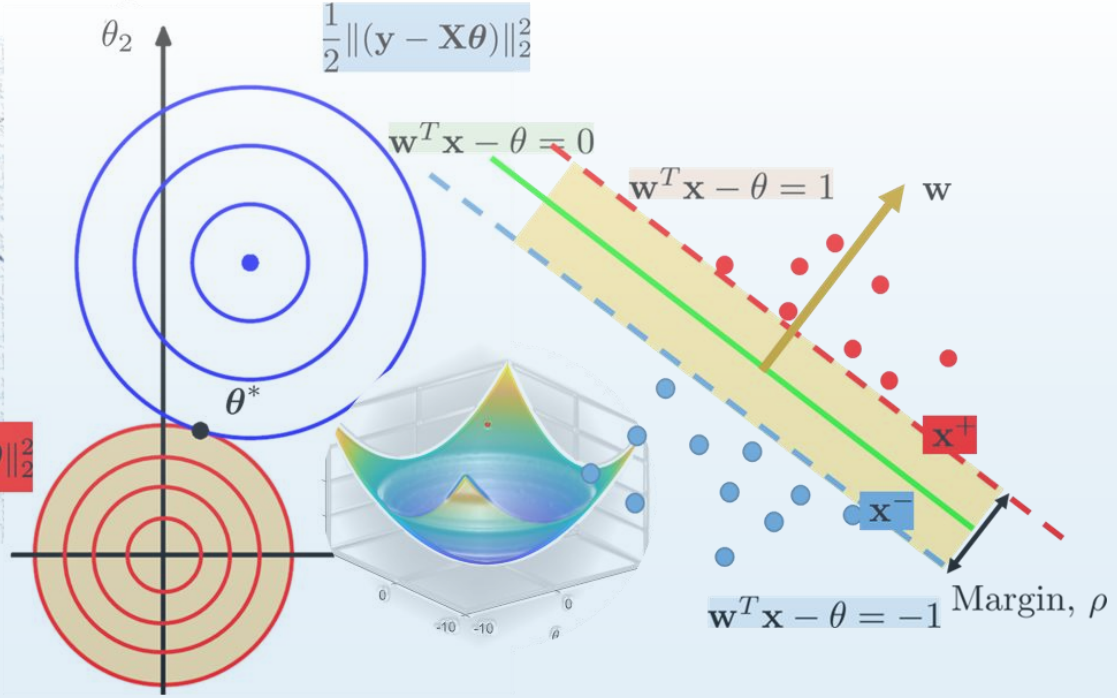




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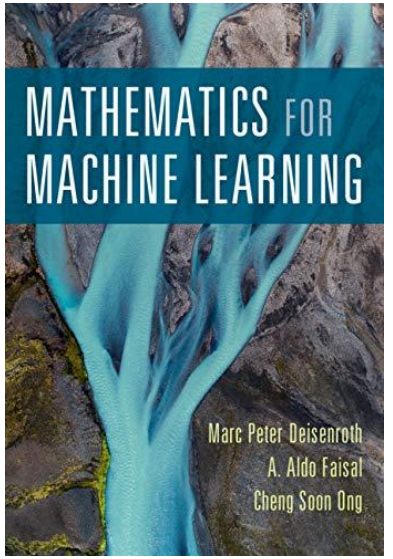
Matrix Rank and Spaces

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Outline

- Matrix Rank
- Column space and Null Space



Chapter 2.6

Matrix Rank

Definition:

- The rank of a matrix is equal to the minimum of number of linearly independent rows and number of linearly independent columns.

Fact: For any matrix $A \in \mathbb{R}^{m \times n}$,

$$\text{rank}(A) \leq \min(m, n)$$

- **Row rank:** Maximum number of linearly independent rows.
- **Column rank:** Maximum number of linearly independent columns.

Key Properties:

- $\text{rank}(A) = \text{rank}(A^T)$
- $\text{rank}(AB) \leq \min(\text{rank}(A), \text{rank}(B))$
- If A is invertible, $\text{rank}(A) = n$

Matrix Rank

Geometric Interpretation:

- Rank represents the dimension of the space spanned by the columns of A.
- For a 3×3 matrix:
  - Rank 3 $\Rightarrow$ columns span $\mathbb{R}^3$ (volume).
  - Rank 2 $\Rightarrow$ columns lie on a plane.
  - Rank 1 $\Rightarrow$ columns lie on a line.

Matrix Rank

Rank and Linear Systems:

For a system $A\mathbf{x} = \mathbf{b}$:

- If $\text{rank}(A) = n$, the system has a **unique solution**.
- If $\text{rank}(A) < n$, the system has **infinitely many solutions**.
- If $\text{rank}(A) < m$, the system is **inconsistent**.

Matrix Rank

Computing Rank via Row Reduction:

The rank can be found by reducing A to its **Row Echelon Form (REF)** or **Reduced Row Echelon Form (RREF)**. The number of non-zero rows in the REF (or pivots in RREF) equals the rank.

1. Write the augmented or coefficient matrix.
2. Apply Gaussian elimination to reduce to REF or RREF.
3. Count the number of non-zero rows (or pivots). This number is the rank.

Example: Compute the rank of

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 1 & 1 & 1 \end{bmatrix}$$

Solution:

$$\begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 1 & 1 & 1 \end{bmatrix} \longrightarrow \begin{bmatrix} 1 & 2 & 3 \\ 0 & 0 & 0 \\ 0 & -1 & -2 \end{bmatrix} \longrightarrow \begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & 2 \\ 0 & 0 & 0 \end{bmatrix}$$

Hence, the rank $= 2$.

Outline

- Matrix Rank
- Column space and Null Space

Column Space and Null Space

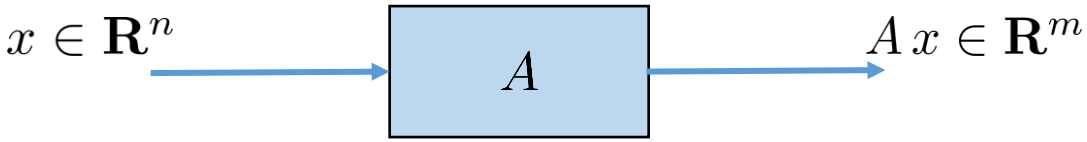
Column Space:

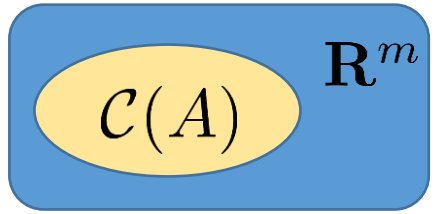
- For a matrix $A \in \mathbf{R}^{m \times n}$, the column space, denoted by $\mathcal{C}(A)$, is the span of the columns of A.
- If $a_1, a_2, \ldots, a_n \in \mathbf{R}^m$ are the columns of A, column space is given by

$$\mathcal{C}(A) = \text{span}(a_1, a_2, \ldots, a_n)$$

$$\mathcal{C}(A) = \{Ax | x \in \mathbf{R}^n\}$$ (all possible linear combinations of columns of A)

- In other words, column space is a linear transformation of every point in $\mathbf{R}^n$, that is,





- Consequently, $\mathcal{C}(A)$ is the subspace of $\mathbf{R}^m$.
- What is the dimension of column space $\mathcal{C}(A)$? Number of linearly independent columns of A=rank(A).

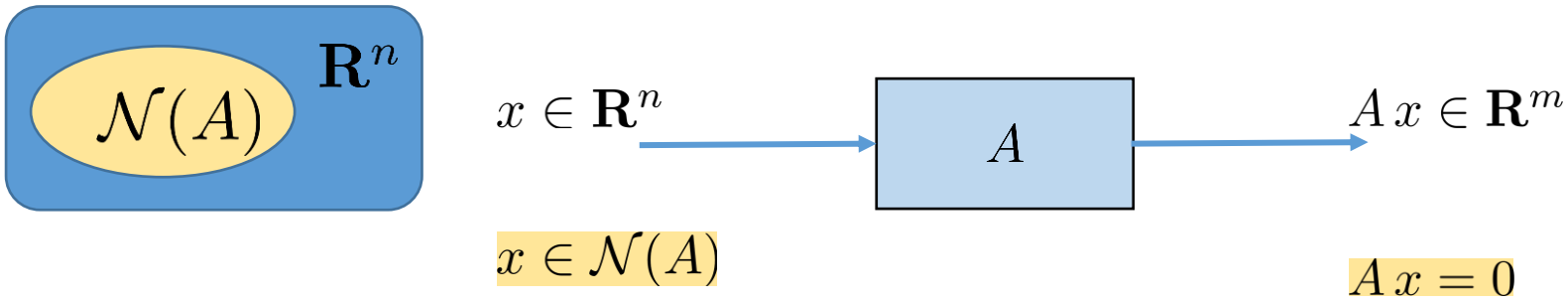
Column Space and Null Space

Null Space:

- For a matrix $A \in \mathbf{R}^{m \times n}$, the null space, denoted by $\mathcal{N}(A)$, is the subspace of $\mathbf{R}^n$ such that

$$\mathcal{N}(A) = \{x \in \mathbf{R}^n \mid Ax = 0\}$$ *(all points that are mapped to zero by matrix A*

- In other words, null space is an inverse linear transformation of $0 \in \mathbf{R}^m$.



- Nullity of the matrix, that is, the dimension of the null-space $\mathcal{N}(A)$ is given by the following rank-nullity theorem (also known as rank+nullity theorem).

$$\text{rank}(A) + \text{nullity}(A) = \text{ number of columns of } A$$

$$\dim(\mathcal{C}(A)) + \dim(\mathcal{N}(A)) = \text{ number of columns of } A$$

Column Space and Null Space

Example:

$$A = \begin{bmatrix} 1 & 1 & 2 \\ 2 & 1 & 3 \\ 3 & 1 & 4 \\ 4 & 1 & 5 \end{bmatrix}$$

- $m = 4$, $n = 3$
- $\mathcal{C}(A)$ is a subspace of $\mathbf{R}^4$.
- $\mathcal{N}(A)$ is a subspace of $\mathbf{R}^3$.

- Note that a third column is a sum of first two columns and therefore number of linearly independent columns is equal to 2.
- Consequently, $\mathcal{C}(A)$ is a 2-dimensional subspace of $\mathbf{R}^4$.